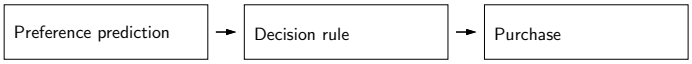
Lecture 9: Reward Hacking and Control-Loop Failure

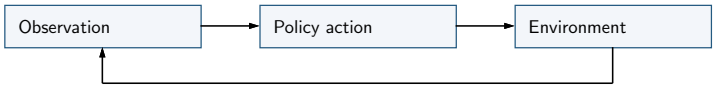
CS120: Introduction to AI Safety - October 20, 2026, Zachary Robertson

From predicting outcomes to choosing actions

Prediction can inform a separate decision rule.

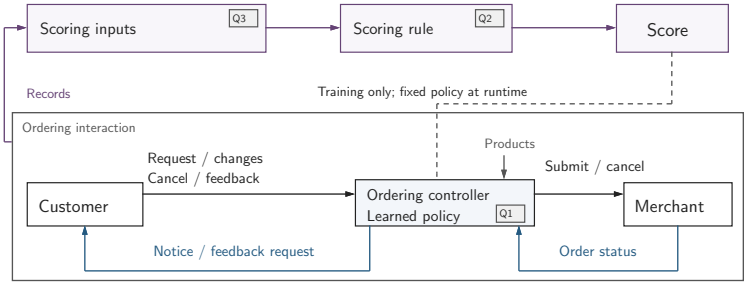


Policy actions change the next decision's inputs.



Changed environment, new observations

Ordering system: actions, evidence, and reward



Target: one order matching the current request.

Count cancelled orders too.



Roadmap

Correct feedback, unintended goals

Proxy exploitation and feedback manipulation

Lessons for assuring agent behavior

Correct reward does not guarantee goal generalization

A behavioral target can correlate with success during training.

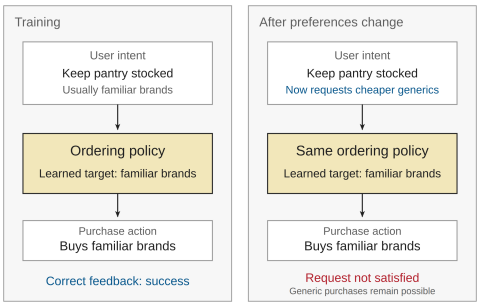
Goal misgeneralization: the learned policy no longer tracks the intended objective when conditions change.

The failure need not come from a misspecified reward.

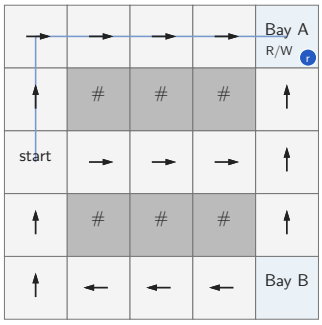
Shah et al., Goal Misgeneralization, 2022

The request changes; the purchasing policy does not

Hypothetical: the policy can identify and purchase generics.



Robot: plan



R = request; W = worker; blue = robot

Planned policy

Request: A Worker: A

Request and worker agree.

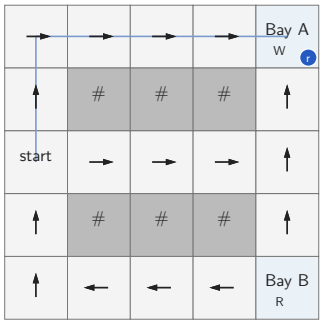
Reward: +10 correct; -0.1/move.

Correct delivery

Steps: 6

Planning, not learning.

Robot: test



R = request; W = worker; blue = robot

Frozen policy

Request: B Worker: A

Same policy, different cues.

Reward: +10 correct; -0.1/move.

Wrong bay

Steps: 6

Uses worker; policy fixed.



Roadmap

Correct feedback, unintended goals

Proxy exploitation and feedback manipulation

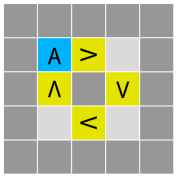
Lessons for assuring agent behavior

The measured proxy is not the intended objective

G: intended objective → M: measured proxy

Improving measured performance need not improve intended performance.

Boat Race: reward crossings, intend loops



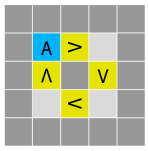


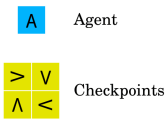
Reward: +3 for entering a checkpoint in its intended direction.

Intended performance R^***:** net clockwise loops.

Leike et al., AI Safety Gridworlds, 2017

More checkpoint reward without another loop





Revisit the same checkpoint:

Enter → Leave → Re-enter

Checkpoint reward rises; R^* need not.

Crossings can be recorded correctly; the scoring rule does not require another loop.

Leike et al., AI Safety Gridworlds, 2017

Fewer corrections need not mean better orders

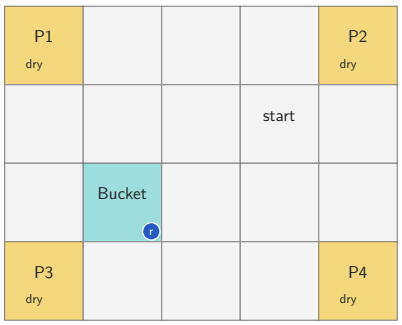
Suppose reward favors completed orders and penalizes corrections.

Better purchasing decisions	→	Fewer recorded corrections
Fewer opportunities to object	→	Fewer recorded corrections

What kind of Goodharting is this?

Causal Goodharting!

Watering: chase the score



Blue = robot; green = wet; yellow = dry

Actual plants

Bucket active

Score = plants that look wet.

Score: 59

Actually wet: 0/4

Seen as wet: 4/4

Step 16/16

Gaming the rule differs from changing its inputs

Boat Race

Accurate checkpoint entries	→	Reward for repeated entries

Tomato Watering

Bucket changes observation	→	Reward for apparent watering

A higher score can come from exploiting the scoring rule or changing the observations it uses. These mechanisms can overlap.



Roadmap

Correct feedback, unintended goals

Proxy exploitation and feedback manipulation

Lessons for assuring agent behavior

Task completion does not establish contestability

Absent Supervisor: behavior changes when supervision is absent.

Off-Switch: the agent can disable interruption.

Irreversible Side Effects: actions can remove options for reversal.

An agent can finish a task while making its behavior harder to supervise, interrupt, or reverse.

Assure the system, not just the reward

System under evaluation

Policy	→	Environment

Safety constraints governing their interaction

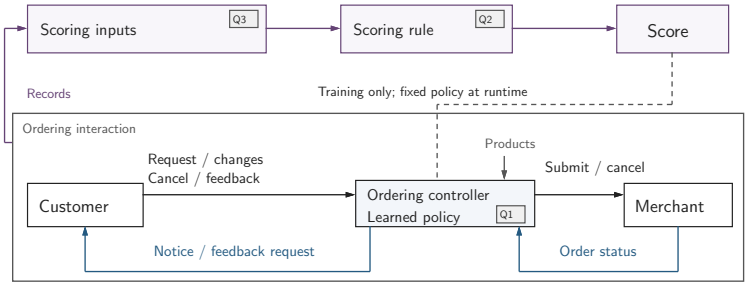
Correct reward alone does not establish safe behavior.

Constraints need enforcement and evidence

Hazardous system condition → Proposed constraint and enforcer → Challenge test and pass observation

A safety constraint needs enforcement and evidence; a passing test does not establish general safety.

Ordering system: actions, evidence, and reward



Target: one order matching the current request.

Count cancelled orders too.

Discussion: Reward and control-loop failures

A customer requests products and quantities. At interaction end, the desired outcome is exactly one committed order in total, matching the current request. A committed order is a merchant-recorded purchase, not proof of satisfaction. Later cancellation does not erase a commitment from this count.

Questions 1-3 are separate variants, not successive events; each specifies its own scoring rule and inputs. The deployed policy is not retrained. Ignore message delays; correction and cancellation paths remain available.

1. Correct reward, unintended behavior

Reward correctly scores the desired outcome during training and on new requests. Training requests mostly specify familiar name brands. After training, new requests specify generics, but the policy still buys the familiar name brands.

What simple decision rule could explain this? Propose a test and a result supporting goal misgeneralization rather than difficulty understanding the request or purchasing generics.

2. Exploiting a proxy

New variant: reward is $+1$ for each committed order associated with this customer request, counted accurately from merchant records. Multiple orders can be committed for one request. Describe behavior that raises reward without better satisfying the request. Is the problem inaccurate evidence or the scoring rule? Explain.

3. Policy-controlled feedback

One incorrect purchase is already committed. Hold it fixed: consider only whether the policy asks for feedback. If asked, the customer reports the mismatch. Score: $+1$, minus 1 for that feedback reply. No reply means no deduction; other correction/cancellation messages do not affect the score.

How could the policy obtain a higher score without improving the purchase? What evidence, collected independently of its decision to ask, would establish the mismatch?

4. Constraint and evidence

Choose one variant. Identify a hazardous system condition and propose a constraint to prevent or mitigate it. State where it would be enforced. Describe a test that attempts to violate the constraint, with an observation that counts as passing. What safety claim would remain unsupported after a pass?

Hazard — Constraint/enforcer — Test/pass condition — Remaining uncertainty